

Link to artifact: <https://claude.ai/code/artifact/57285549-6c49-44d8-8567-de3569166d7e>

The context

Most caffeine apps are calorie-counters wearing a different hat: log a drink, watch a number go up, feel vaguely bad. They answer *how much*.

“Half-life” answers a different question: *when*. **As you are building, assume the following: 200 mg of caffeine has a half-life of roughly 5–6 hours, so a 4pm cold brew still has 80 mg circulating in your bloodstream at 11pm.** The interesting insight isn't your daily total; it's the decay curve and where your bedtime falls on it.

You're building that app. The attached prototype is a working directional reference, not a spec. Treat it as the strongest existing articulation of the idea, and improve on it where you can defend the change.

Your core task

Ship a working version of the Half-life app.

At **minimum** it must:

1. Log a drink in one tap
2. Pull sleep, steps, etc. from Apple Health.
3. Display the correlations between the two cleanly, and beautifully.
4. Build a profile for a user of their longitudinal caffeine drinking habits and how they might be affecting sleep, heart rate, etc.

Design decisions matter as much as the build. If you cut something from the prototype, say why. If you add something, defend it.

Bonus: text the app. Include text-message logging. A user should be able to text a number and have the drink logged on the app, or ask the text message when the optimal time to sleep would be. The user should be able to ask questions about their data and get answers that feel more thoughtful than a correlation coefficient

Scope and constraints

- **The platform is yours to choose.** Native iOS is the obvious fit given the Health integration and the visual language. React Native, or a web app with a documented Health stand-in, is acceptable if you tell us what you traded away.
- **One user, one device.** No accounts infrastructure, no sync, no social features.
- **Fake data is fine and encouraged.** Seed 2-4 weeks of plausible history so the Patterns surface has something to say. Say clearly what's seeded vs. live.
- **Don't build:** notifications infrastructure beyond a demo, a drink database beyond ~15 common items, settings screens for their own sake, or anything resembling a streak.
- **AI assistance is allowed.** Use whatever tools you'd use on a real job. We care about the result and your reasoning, not your keystrokes.

What to submit

1. **The working thing:** repo link plus a TestFlight build. Include setup instructions that work on a clean machine.
2. **A short write-up, max 2 pages.** We want:
 - The one or two decisions you're proudest of, and what you rejected to get there
 - What you cut, and why it didn't earn its place
 - How you handled the half-life math and what you'd fix given another week
 - If you attempted the bonus: your parsing strategy and its failure modes
 - If you had more time, how you would have handled data privacy
3. **Anything you made along the way** that shows the thinking; sketches, dead ends, notes. Unpolished is fine.
4. If you used AI, **share the time-stamped scripts of your AI interactions.**

Send everything to team@femmli.com by 5pm ET on Friday, September 18. If something's blocking you before then, email us; asking a good question is not a mark against you.

How we'll evaluate

What we're looking for

Judgment	Did you protect the single question, or did the app sprawl? Are the cuts principled?
Craft	Does the decay curve feel alive? Do the numbers land with the weight the type gives them? Does one-tap logging actually take one tap? Is the design beautiful?
Honesty	Does the app overstate what it knows about your sleep? Confidence language matters here more than the correlation itself.
Communication	Can we understand your reasoning without you in the room?
Bonus	Attempted thoughtfully (even partially) counts for more than skipped. Attempted badly at the expense of the core does not.

A note on effort

One week means one week of *part-time* work. We're assuming 10-15 hours, not 60. If you find yourself grinding, cut the scope and tell us what you cut. Knowing when to stop is part of what we're evaluating.